



Lecture 11: The Default Export

INTERVIEW QUESTION

What is the default export of a compiled Svelte component file when imported into a JavaScript module?

When you write a `.svelte` file, you are writing a declarative mix of HTML, CSS, and JavaScript. The browser cannot natively understand this format. However, when the Svelte compiler processes your file, it transforms it into a highly optimized, standard JavaScript module.

The most important part of this transformation is what the compiler exposes to the rest of your application: **a default export**.

Svelte 5: The Setup Function

In Svelte 5, the default export of a compiled component is a plain JavaScript function (often referred to as a "setup" or "render" function).

When you import a component into another file, you are importing this compiled function. The function's job is to wire up the reactive signals (Runes), attach event listeners, and mount the specific DOM nodes required by that component.



The screenshot shows a code editor window titled "App.svelte". The code is as follows:

```
1 <script>
2 import ProfileCard from './ProfileCard.svelte';
3
4
5 <ProfileCard />
```

Annotations in the image:

- A callout box pointing to line 2: *Imports the compiled default function.*
- A callout box pointing to line 5: *You rarely invoke this function manually. Instead, Svelte's compiler handles the invocation when it sees the markup below.*
- A callout box pointing to the opening tag of line 5: *Svelte automatically invokes the imported function here*

KEY TAKEAWAY

A `.svelte` file is just the authoring format. The compiled output is a standard JavaScript module whose `default` export is a function that sets up the component's reactivity and DOM.

INTERVIEW TIP

To impress the interviewer: Highlight the architectural shift between Svelte 4 and Svelte 5. In Svelte 4, components compiled into JavaScript **Classes**, which had to be instantiated with `new Component(...)`. In Svelte 5, the compiler outputs plain **Functions**. This shift eliminates the memory overhead of class instantiation and aligns perfectly with Svelte 5's fine-grained, signal-based reactivity model, resulting in smaller bundles and faster execution.

Summary

The default export of a compiled Svelte 5 component is a standard JavaScript setup function. This function contains all the hardcoded DOM instructions and reactivity wiring necessary to bring your declarative `.svelte` markup to life in the browser.

	<code>.svelte</code> FILE (Authoring Format)	COMPILED JS MODULE (Browser Output)
Format	Declarative HTML/CSS/JS mix	Standard JavaScript Module
Default Export	(Not a valid JS module)	A Setup FUNCTION
Execution	Needs to be compiled	Invoked natively by the browser